

Name:-Suryansh Jaiswal

College:- Indian Institute of Technology Patna

Round 1 of interview (DSA ROUND)

In this round the interview asked me two dsa Questions easy to medium:-

1 <https://leetcode.com/problems/number-of-islands/>

2 <https://leetcode.com/problems/3sum/description/>

(expected time complexity $O(n^2)$).

Then he asked me to ask question if i have any and then we had light discussion for 10 min and he left overall interview went well was able to solve both question in about 50 min and 10 min was little discssion with him during 50 min he asked me some theoretical dsa question also like difference

between complete binary tree and perfectly balanced binary tree and space complexity of spanning a perfectly balanced tree and then obviously there were a lot of questions based on the time and space complexity on the about 2 questions which he asked and was very much oriented towards reducing the space complexity as much as possible.

Round 2 of interview (ML BREADTH AND DEPTH ROUND)

After giving the first Round of interview on the same night I got the mail saying that my round 2 is scheduled tomorrow and I was not prepared for that but though I had good understanding of theoretical ML so I just needed to brush up my concepts I remember I didn't sleep whole night until my interview got over next day which was from 10 AM it went

for around 2 hrs they first started with talking about my project and then deep dived into the ML concept covering Boosting, Baging, Bias-variance trade off, ensemble learning, supervised learning, loss function, neural networks, entropy, gini index, decision tree, baye's theorem, unsupervised learning(clustering), K-means, KNN, RNN, LSTMS, TRANSFORMERS and many more few of the practical situation based question were also there like what algorithm or model would you use in this situation and why, then they asked me to implment a whole of neural network , softmax loss function and to write code of mean squared loss in 3-d matrix the interview was very rigourous went for almost 2 hrs but thankfully i was able to answer 95% of the questions they asked correctly and sometime when i went in worng direction the interviewer used to guide me also he was a nice guy who made me very comfortable in the

interview and It was much like a healthy discssion rather than a interview.

Result :Selected (you can connect me on [linkedin](#)).

Name: Yaswanth Kumar

College: Vellore Institute of Technology, Chennai

Round 1 of interview (DSA ROUND)

**In this round the interviewer asked me two dsa Questions
easy to medium:-**

- 1) Delete a Node from linked list without head pointer**
- 2) Most frequent character in string grouped by characters**

For 2nd question I started with an $O(n^2)$ solution for the second question, but the interviewer challenged me to optimize further. Initially, I implemented binary search, but I hit a roadblock. Thankfully, the interviewer provided helpful guidance, and I ultimately landed on an efficient $O(26 \cdot \log(n))$ solution. This round went well, and I feel confident moving forward.

Round 2 of interview (ML BREADTH AND DEPTH ROUND)

A week after the first round, I received an invitation for the second round. The interview began abruptly with no introductions. Two interviewers were present, one focusing on ML Breadth and the other on ML Depth, each for 30 minutes.

The ML Depth interviewer delved into my project, questioning my choice of model and prompting me to suggest potential new features in and solutions. He also tested my understanding of core concepts like cross-entropy, neural networks, SGD, and others.

The ML Breadth interviewer shifted the focus to the broader ML life cycle and real-world scenarios where ML algorithms could be applied. He also inquired about difficulties faced with the dataset and how I would overcome them.

The interview concluded with questions from me, followed by the interviewers' assurance that the HR team would be in touch.

Result: Not Selected

You can connect me on [LinkedIn](#)

Name:- S.V.S.Sruti

College:- Indian Institute of Technology Patna

Round 1 (DSA)

In this round the interviewer asked me three DSA Questions easy to medium:-

1) Implement Deque

(<https://www.geeksforgeeks.org/implementation-deque-using-circular-array/>)

I implemented using array.

Follow up questions -

What is this type of array called? (circular array)

Which other data structure can be used to implement deque ?
(doubly linked list)

He also kept asking about the explanation of lines in between.

2) Given an array arr[] of size N. Find the sum of the contiguous subarray with the largest sum.

(<https://www.geeksforgeeks.org/largest-sum-contiguous-subarray/>)

First, he asked me to say the brute force approach
Then, he asked me to optimize the code which I did using Kadane's Algorithm.

3) Maximum sum such that no two adjacent elements are adjacent

[\(https://www.geeksforgeeks.org/maximum-sum-such-that-no-two-elements-are-adjacent/\)](https://www.geeksforgeeks.org/maximum-sum-such-that-no-two-elements-are-adjacent/)

I first gave him the naive approach that is using recursion.
Then with a little optimization, that is, top-down method
And finally, the bottom-up method.

He, as well, asked the Time Complexities for each code.

Round 2 (ML Depth and Breadth)

The interviewer started with a brief introduction of himself and then he asked me to introduce myself. After the introduction, he asked me "What is that thing that you could not speak about in your previous rounds?" I straight away started talking about my project. I, first, gave a brief introduction about the project. Then, he asked, "What is the thing that you like about ML/DL/projects?" . Then, I spoke about the neural networks, the functioning of it, etc. Gradually, I started to talk about the details of my project. He asked me, "where did you get the dataset from?", "Did you label all the images in the dataset by yourself?" And then there was deep discussion about the dropout layer, the functioning of it for which you need to cover answers for the following questions -

"What is the main function of the dropout layer?" (prevents overfitting)

"How does it actually work?" (few neurons in each layer will be

dropped out, etc)

“How does it work during training and testing?” (explain about how the weights get multiplied with the parameter p during training and testing and why?)

“Why is the dropout layer turned off during testing?”

While I was talking about how the weights get multiplied with $(1-p)$ during testing, he asked me “Can I know why the weights get multiplied with $(1-p)$?”, “What will happen to the activation functions if they don't get multiplied with $(1-p)$?”

The discussion on the dropout layer took around 30 min.

After this, he asked me about the performance metric I used in my project and why. “What are the other performance metrics you know and where can you use them?” Then, he gave me three situations before COVID-19 where wearing a mask isn't considered necessary, during COVID-19 where wearing a mask is a necessity and after COVID-19 where again it is back to normal. He asked “what performance metric will you use in these situations?”

“How will you change the beta value in the f1 beta score if you focus more on precision?”

He kept cross-questioning in between for the answers I gave him. This went on for the next 20 min.

And then he asked a simple probability question “There are 7 coins. Each coin has two sides - head and tail such that the probability of getting a head is 0.9. What is the probability of getting exactly 5 heads and at least 5 heads?”

“How many ways are there to get all 7 heads?”

He asked me to write the answers for these in that chat.

And then finally, he concluded by asking “Do you have any questions for us?”

Overall, it was a great experience!

Result:- Selected

Name:- Tejas Alisyam

College:- BITS Pilani Hyderabad Campus

Round 1 : DSA

I was asked a behavioral question

-> what is the biggest decision you have taken in ur life

Be prepared with all the 16 leadership principles and frequently asked questions which can be easily found online

Two dsa questions were asked

- 1) Very similar question to the **Next Greatest Element** (monotonic stack)

[leetcode](#)

It was more like previous greatest element, but the idea is the same. I forgot about the edge case, but got the idea when he changed the test case which covered the edge case

- 2) **print all the anagrams when a vector of words is given**

Method: Sort every word and place them in unordered_map

In this, the main task was to sort each word in $O(n)$. (Ans: as each word contains only 26 characters, create an array of size 26 and increase the count of all the characters present in the word. Then traverse the array again from the beginning which will give u the sorted word)

Round 1 lasted for about 1 hr, got the mail for the second round that night itself,

ROUND 2 : ML Breadth and Depth Round.

Two applied scientists were the interviewers

Part 1: depth round

It started off with me explaining about my projects. My projects were related to LSTM and CNN.

They mainly looked at the understanding I have about these topics and why specifically these were used in my projects

I was continuously asked for 30 min about my projects going into deep depth about everything I said.

Second part: breadth round

A few questions asked me were :

- 1) Bagging vs boosting, what all boosting techniques do you know, when to use which ensemble method. Bagging was a random forest.

- 2) Bias, variance
- 3) PCA
- 4) Dropout
- 5) About different CNN architectures,(I only knew resnet, so said only about it)
- 6) Unsupervised learning, different types of clustering
- 7) Explain k mean clustering, what is the loss function used

My second round lasted for around 55 min.

PREPARATION TIP :

-> BREDTH ROUND: Use these questions and also try to scrape questions from the internet, u can find around 100-150 questions, and they will cover majority of the topics.

-> DEPTH ROUND: Be honest with your projects. Have a deep mathematical understanding of every algorithm used.

You can connect with me at [LinkedIn](#) .

Verdict : Selected